Cardiovascular Impact of Consumption of Sorghum Based Alcoholic Beverage (*Burukutu*) in Adult Male Wistar Rats

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ABSTRACT

Alcohol production companies and business are making so much of profit from increased products demand but the economic constrained have seek to depend on the locally made version. Cardiovascular diseases are on the increase and lipid profile markers are suitable indicator for the determination of cardiovascular health. This study attempts to elucidate the impact of sorghum based alcoholic beverage (*Burukutu*) on cardiovascular health through changes in serum lipids in rats. Thirty male Wistar rats weighing 80-100g were divided into 6 groups; a control, groups treated with 5% *Burukutu* every other day, weekend and those treated with 1%, 5% and 25% *Burukutu* daily. At the end of 14 days treatment, blood samples were collected for assessment of serum lipid profile. The data was subjected to ANOVA at $p \leq 0.05$ significance. The results showed that serum total cholesterol, triglyceride, high density lipoprotein (HDL) and low density lipoprotein (LDL) levels in rats administered *Burukutu* were not significantly different compared with the control. However, total cholesterol, triglyceride, and high density lipoprotein levels were likely decreased following weekend ingestions compared to every other day with LDL increased following weekend ingestions compared to every other day. These findings suggest that ingestion of *Burukutu* may not affect the functionality of the cardiovascular system in rats. However, every other day ingestion and weekend ingestions are most likely to have significant impact on the cardiovascular system considering their effect of triglyceride and total cholesterol and LDL respectively.

Keywords: Sorghum based alcoholic beverages, Burukutu, Cardiovascular healt, Rats.

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1. INTRODUCTION

Alcohol and traditional alcoholic beverages feature importantly in many traditional Nigerian cultural groups and their consumption has a long history and use is common amoung all cultural groups in the country [1]. Traditional alcoholic beverages such as *burukutu, pito, emu funfun, ogogoro* and *agadagidi* are among the most available psychoactive substances in the country [2].

Burukutu is a popular alcoholic beverage produced mainly from the grains of guinea corn of the species *Sorghum vulgare*and *Sorghum bicolor* involving steeping, germination, fermentation and maturation and resulting to a cloudy alcoholic beverage [3]. The product is said to contain almost all essential amino acids in required proportion except cystine and tryptophan which are being completely destroyed by heat during boiling [3].

Since burukutu is an alcoholic beverage, it is likely that all the health risks commonly associated with overconsumption of alcoholic beverages [4] may also apply to the misuse of burukutu. Documented data on the effects of burukutu is few and thus, no much research has been done to understand how these locally brewed beers affect some body organs and system. The present study is therefore designed

to investigate the effect of laboratory processed burukutu on lipid profile in rats. The results of this study will help to provide information on the probable effects of this locally brewed beer on serum lipid profile of consumers and the general public.

2. MATERIALS AND METHODS

2.1. Materials: Sorghum (*Sorghum bicolor*) was locally purchased in market square from Hausa Quarters, the local market in Ekpoma, Edo State. The animals (adult male Wistar rats) were obtained from the Animal Holding Unit of the Faculty of Basic Medical Sciences, College of Medicine in Ambrose Alli University, Ekpoma. Rat chow (grower's mash) was palletised chicken feed (Grower) produced by Grand Cereals Ltd, Jos Plateau State, Nigeria and was purchased in Ekpoma, Edo State, Nigeria. All other chemicals were of analytical grade obtained from the physiology laboratory, Ambrose Alli Uiniversity.

2.2 Preparation of the Sorghum based alcoholic beverage (*Burukutu*)

The production of Burukutu in this study was done following steps categorized broadly into malting, fermentation of mash and the maturation process as in the method described previously by Wedajo Lemi [5] with few modifications as in our prevous paper [6].

2.3. Experimental and grouping: Rat used in this study were giving standard care as stated in the guidelines and regulations for the use of animals in scientific experiment. They were housed in a plastic cage (60 cm x 30 cm x 30 cm) under a 12/12 hours light/dark cycle for two weeks of acclimatization and fed rat chow and water *ad libitum.* The rats (N = 30; weighting between 100±20g were divided into 6 groups with group A as control. Group B received 5% *burukutu* mixed with their diet at the beginning of treatment and every other day for the 14-day study. Group C received 5% *burukutu* moxed in their diet at the beginning of treatment and every weekend (Fridays, Saturdays and Sundays) for the 14-day study while groups D, E and F received 1%, 5% and 25% *burukutu* mixed in their diet daily for the 14 days study. The required percentage of *burukutu* was mixed with the rat chow daily at 9.00am and given to the rats to eat *ad libitum.* As in Oyakhire *et al.* [6].

2.4. Sample collection: At the end of the experiment, blood sample was collected following standard laboratory procedures using 2mL syringe. The blood sample was centrifuge at 3000 rpm and the serum obtained for lipid profile using commercial kits (Randox Laboratories kits, United Kingdom) as per the procedure described in the kits. The concentration of low density lipoprotein (LDL) was calculated from the Fiedewald's [7] formula: LDL = TC – HDL – TG/5. Where TC = Total Cholesterol, HDL = High Density Lipoprotein Cholesterol, TG = Triacylglycerol.

2.5. Statistical analysis: The mean ± standard error of mean was determined and the one-way ANOVA statistical test was performed using SPSS version 20 software. The value $P < 0.05$ was considered statistically significant. Results were presented as mean ± Standard deviation using suitable tables and charts.

3. RESULTS

Figures 1 present serum total cholesterol, triglyceride, high density lipoprotein and low density lipoprotein levels following ingestion of *Burukutu* in adult male Wistar rats. Serum total cholesterol, triglyceride, high density lipoprotein and low density lipoprotein levels following ingestions of *Burukutu* for 14 days were not significantly alter compared with control group. However, ingestion of *Burukutu* on a weekendly base presented lower serum cholesterol, triglyceride and high density lipoprotein levels compared to every other day ingestion group with the opposite occuring with serum low density lipoprotein cholesterol.

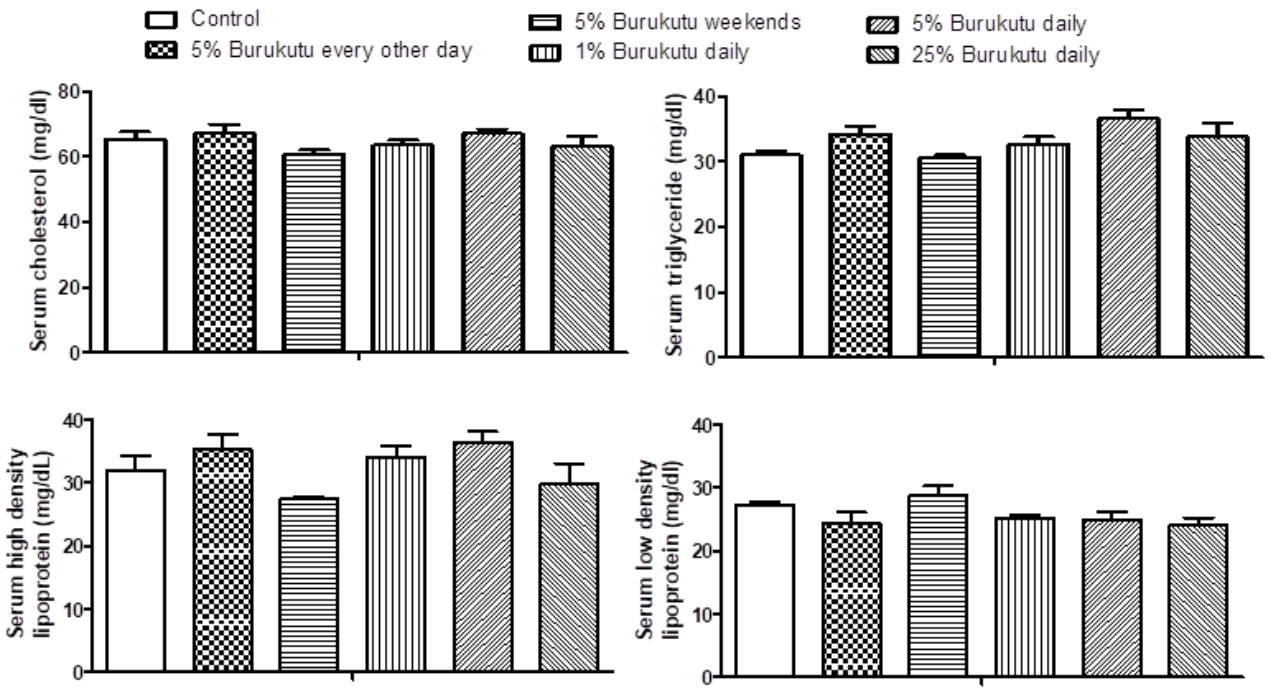


Figure 1. Serum (a) total cholesterol, (b) triglyceride, (c) High and (d) low density lipoprotein-cholesterone levels in rats treated with locally made sorghum based alcoholic beverage.

Values are means±SEM, n = 5, No statistically significant different compared with untreated control at $p \leq 0.05$.

4. DISCUSSION

Dyslipidemia is a group of plasma lipid and lipoprotein abnormality that is metabolically associated, and it is categorized by low HDL and increased LDL, TGs, and total cholesterol levels [8]. Elevated serum lipids, including cholesterol and triglycerides—a condition known as hyperlipidemia—significantly increase the risk of atherosclerotic CVD. Elevated serum total cholesterol may exacerbate coronary heart diseases [9]. Dyslipidemia has implications in the cardiovascular system, producing pathologies as vascular coronary disease and atherosclerosis [10]. This study was therefore undertaken to investigate the changes of some serum lipid profile variables in response to ingestion of locally made sorghum based alcoholic beverage (*Burukutu*) in adult male Wistar rats.

The total cholesterol, triglyceride and HDL concentrations increased non-significantly in the groups treated with *Burukutu* compared to the untreated control; especially in the vary doses and every other day treated groups. This is an indication that chronic and higher dose ingestion of burukutu may alter lipid profile and exacerbate coronary heart diseases. In this regards, a study has documented that long-term consumption of *burukutu*, can negatively affect lipid profiles by inducing oxidative stress and damaging the liver [11]. A study on rats found that burukutu consumption significantly elevated levels of lipid peroxidation products, indicating toxicity and oxidative stress [11]. *Burukutu* has been reported to produced significant change in the activities of liver CYP2E1 [11]. This increase may enhance ROS generation (superoxide anion, hydroxyl radicals, hydrogen peroxide, and hydroxyethyl radicals), which could lead to lipid

peroxidation, oxidative protein damage, and DNA oxidation [12-14].

For LDL, the concentration was observed to decrease non-significantly in the *burukutu* treated groups compared with the untreated control. The decline of LDL cholesterol in *burukutu* administered rats may be due to the fact that alcohol consumption generates an increasing activity of paraoxonase. Paraoxonase confers to HDL antioxidizing properties by making possible the hydrolysis of LDL oxidized phospholipids [15]. Cholesterol level reduction, especially LDL cholesterol, which could be a favorable dynamic for a healthy heart, a situation that may have been accomplished by either diet, *burukutu* or both.

CONCLUSION

The present study showed that dyslipidemia may not be presence following 14 days ingestion of locally made sorghum based alcoholic beverage (*Burukutu*) in adult male Wistar rats. However, the reduring in HDL following weekend and 25% daily ingestions are possible detrimental effect on serum lipid profile and hence, the need for further studies.

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